HCIVR: An AI-Personalized Virtual Reality Meditation Environment for Stress Reduction

Final Project Report (CS 449 / 549 – Human Computer Interaction)

Ali Furkan Dumlu 31201 (Unity implementation, interaction integration)
Lara Çelen Gürbüz 30918 (UI design, interaction feedback components)
Boray Saydam (backend codes, system connectivity)
Mükremin Berkay Can 37999 (AI pipeline design, adaptive guidance flow)
Bahattin Taşçı 31076 (survey design, qualitative & quantitative questions)
Kadir Şen 28864 (participant recruitment, user testing, data collection & analysis)

January 2026

Project short name: HCIVR
Public repository and materials:
`https://github.com/kadir-sen/SU-CS449-549-Final-Project-HCIVR.git`

1 Introduction

Digital stress and burnout have become a routine part of everyday life, especially for university students who navigate constant deadlines, dense schedules, and continuous online exposure. Although meditation and wellness applications are now common, many of them are built around static content and assume that users can easily "switch into" a calm mental state by following instructions. In practice, this assumption often fails precisely when users need support the most: during moments of fatigue, anxiety, or cognitive overload.

HCIVR was developed as a response to this gap. The project investigates whether an immersive VR environment combined with adaptive, AI-supported guidance can help users relax with less effort. Rather than treating relaxation as a skill users must actively perform, HCIVR treats it as an experience that can be scaffolded through interaction design: clear feedback, minimal decision-making, and a coherent end-to-end user journey.

From an HCI perspective, the core contribution of HCIVR is not a technical pipeline by itself, but a user-centered design and evaluation of an adaptive interface in a sensitive domain. The project therefore emphasizes the Four Pillars of Design framework: (1) clearly defined user expectations and context, (2) theory-grounded process and documentation, (3) implementation as a realized interface artifact, and (4) evaluation with real user interaction.

2 Pillar 1: User-Interface Requirements (User Expectations & Context)

2.1 Target users and domain context

The primary target users are Sabancı University students experiencing moderate everyday stress. The system was intentionally designed to be approachable for users who are new to VR. In our study, the majority of participants (9 out of 14) were first-time VR users, which made learnability, clarity, and perceived safety central requirements rather than optional qualities.

The domain context is digital wellbeing and stress reduction. This context imposes two constraints that shaped our requirements early: first, stressed users have limited patience for confusing interfaces; second, users are particularly sensitive to uncertainty and loss of control during emotionally loaded interactions. A relaxing system must therefore avoid ambiguous states and must communicate what is happening in a way that feels reassuring rather than intrusive.

2.2 User need and problem framing

We observed a recurring mismatch in many meditation tools: users are expected to choose settings, interpret instructions, and sustain focus, even though stress often reduces the very capacity required for these actions. In other words, the interface asks users to do "work" in order to relax.

HCIVR addresses this by aiming for a low-effort interaction model: users should be able to start quickly, provide a minimal emotional input, and then be guided through a coherent session that provides closure. The interface is not designed as a feature showcase, but as a journey with clear stages and predictable transitions.

2.3 Scenario and user journey

A typical usage scenario involves a student finishing an intense academic day and seeking a short mental reset before continuing other tasks. In this setting, the system must (i) reduce setup complexity, (ii) clearly communicate that the system is responding correctly, and (iii) avoid requiring repeated manual decisions that may increase cognitive burden.

The intended user journey can be described as a four-stage task flow: (1) session initiation, (2) expression of current mood or need, (3) adaptive guidance and environment behavior, and (4) explicit confirmation that the session has concluded successfully. This user journey functioned as a grounding design artifact and guided our interface structure more strongly than individual aesthetic decisions.

2.4 HCI significance: Reducing cognitive load through adaptive design

A central HCI contribution of HCIVR is the reduction of decision-making burden during emotionally sensitive interactions. Without AI-supported adaptivity, users would face sequential choice screens requiring them to select from multiple options before beginning their meditation session. This design pattern, while common in configuration interfaces, directly conflicts with the goal of reducing cognitive load for stressed users.

To illustrate the HCI significance of our adaptive approach, we apply Hick-Hyman Law, which quantifies decision time as a function of the number of choices. Hick-Hyman Law states:

$$RT = a + b\log_2(n+1) \tag{1}$$

where RT is reaction time, n is the number of choices, and a and b are empirically determined constants. For interaction design, typical values are approximately $a = 50$ ms and $b = 150$ ms (Card, Moran, & Newell, 1983).

Initial design scenario (without AI adaptivity): Users would encounter two sequential selection screens. The first screen would present 8 environment choices (forest, ocean, mountain, desert, space, garden, rain, sunset). The second screen would present 8 emotional state options (happy, stressed, anxious, sad, energetic, tired, calm, frustrated).

For the first choice screen (8 environments):

$$RT_1 = 50 + 150\log_2(8) = 50 + 150 \times 3 = 500 \text{ ms} \tag{2}$$

For the second choice screen (8 emotional states):

$$RT_2 = 50 + 150\log_2(8) = 50 + 150 \times 3 = 500 \text{ ms} \tag{3}$$

Total decision time before meditation begins:

$$RT_{total} = RT_1 + RT_2 = 500 + 500 = 1000 \text{ ms} \approx 1 \text{ seconds} \tag{4}$$

This calculation represents only the raw cognitive processing time and does not account for visual search time, reading comprehension, or hesitation due to uncertainty about which choice best matches the user's current state. In practice, observed decision times for such interfaces are often 2–4 times longer than Hick-Hyman predictions when users experience decision paralysis or choice overload.

AI-supported design (current implementation): Users provide a single natural language voice input describing their current emotional state. The system interprets this input and automatically selects the appropriate environment and guidance parameters. Decision time is effectively reduced to near-zero from the user's perspective, as the cognitive burden is transferred to the AI adaptation layer.

Figure 1 illustrates the two sequential choice screens that would have been required in a non-adaptive design. By eliminating these decision points, HCIVR directly addresses a core HCI principle: interfaces should minimize unnecessary cognitive load, especially in contexts where users are already experiencing stress or reduced attentional capacity.

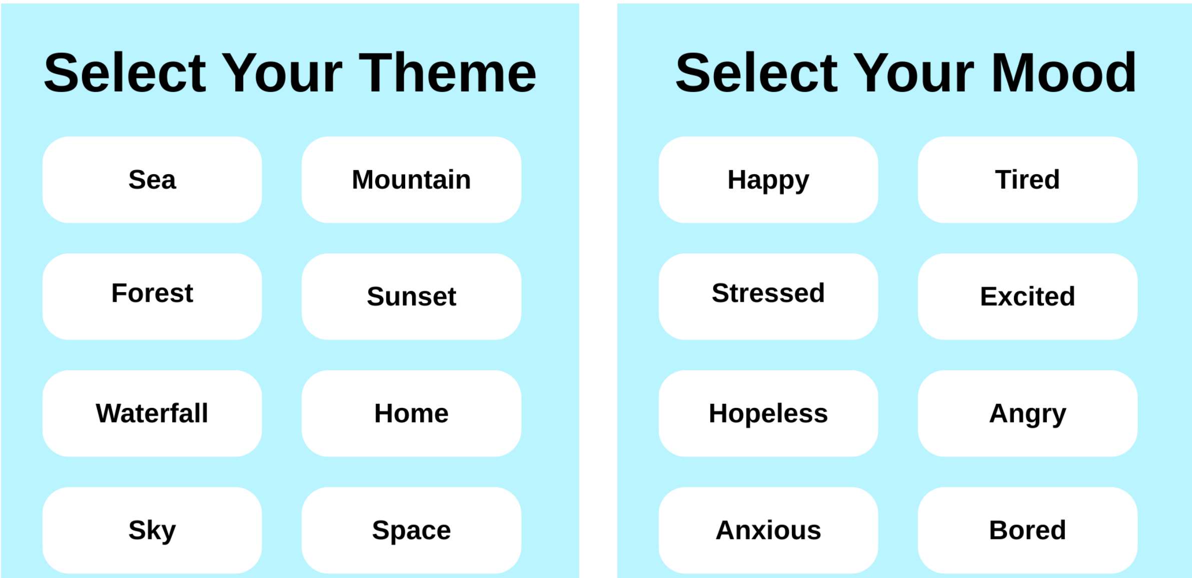

Figure 1: Initial design concept showing sequential choice screens that would have imposed cognitive load. Left: environment selection (8 choices). Right: emotional state selection (8 choices). These screens were eliminated through AI-supported adaptivity.

This design decision is particularly important for first-time VR users, who already face cognitive overhead from spatial navigation and novel interaction modalities. Our evaluation results confirm that reducing pre-session decision-making contributed significantly to perceived comfort and ease of use.

3 Pillar 2: Guidelines, Documents & Design Process

3.1 Theoretical grounding

We grounded our design decisions in core HCI principles that are especially relevant in high-uncertainty or emotionally sensitive contexts. Norman's Gulfs of Execution and Evaluation provided a practical lens: users should not struggle to map intentions ("I want to calm down") to actions, and they should not struggle to interpret the system state once they act. Related to this, visibility of system status was treated as a primary design rule, because silent or ambiguous processing states can easily trigger anxiety.

We also treated cognitive load reduction as a central design goal. Rather than presenting multiple configuration choices or complex control structures, we aimed to keep the interface lightweight and to externalize complexity into the system logic. Finally, because HCIVR includes adaptive behavior, we explicitly considered trust calibration: adaptive responses should be understandable and predictable, rather than perceived as opaque or unexplained automation.

3.2 Low-fidelity design and iterative refinement

Before implementation in VR, we developed low-fidelity design artifacts to reason about interaction flow, feedback timing, and user expectations independent of technical constraints. These artifacts included interaction flow sketches and interface wireframes, primarily created in Figma.

The low-fidelity designs focused on identifying moments where users might experience uncertainty, such as during voice input or system processing. Insights gained at this stage directly informed later implementation decisions, including the use of guidance text to confirm that the system is listening, a visible loading indicator during processing, and an explicit success screen at the end of each session. This iterative refinement ensured that theoretical principles were translated into concrete interface behaviors.

3.3 C4 container-based development process

To maintain a traceable and systematic design process, the system architecture was structured using a C4-inspired container-level perspective. At this level, HCIVR consists of three primary containers: the VR user interface, the AI adaptation logic, and the backend connectivity layer.

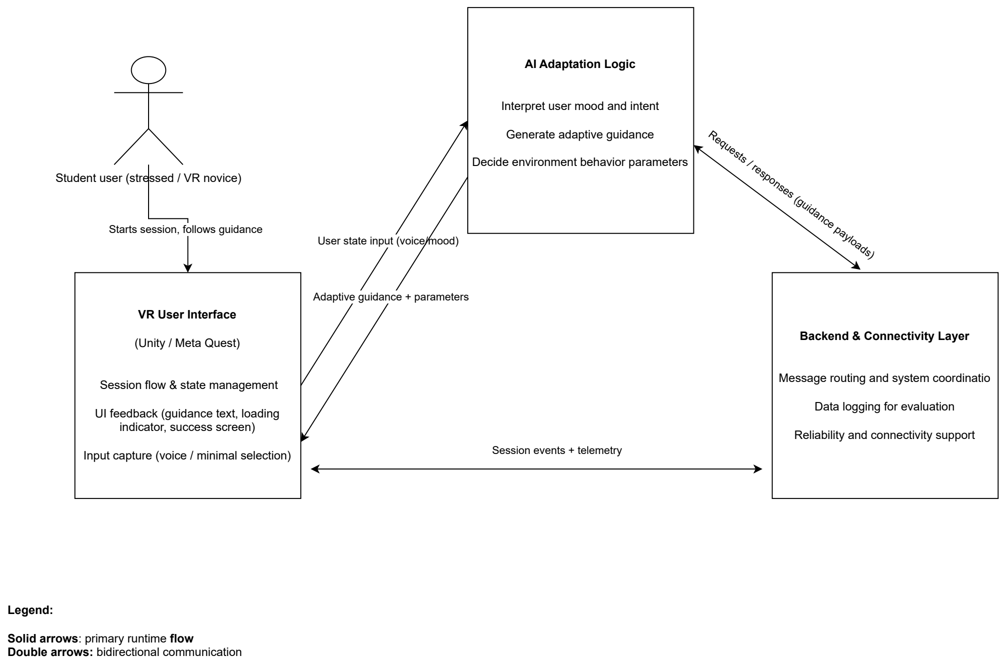

Figure 2: C4-inspired container-level view of the HCIVR system.

The VR user interface container, implemented in Unity, manages session flow, captures user input, and communicates system status to the user. The AI container interprets user input and generates adaptive guidance and environment parameters. The backend container ensures reliable communication between components and supports data logging for evaluation purposes. This separation helped align conceptual design decisions with implementation artifacts.

3.4 Team workflow and traceability

The project progressed with a role distribution that supported parallel development while maintaining a coherent HCI narrative. This structure allowed design intent to be traced through low-fidelity artifacts, implementation decisions, and ultimately to observed user experience during evaluation.

4 Pillar 3: User-Interface Software Tools (Implementation)

HCIVR was implemented in Unity and deployed on VR hardware (Meta Quest). Unity was chosen because it supports rapid iteration on interactive VR experiences and enables consistent control over session flow, transitions, and interface overlays. Figma was used during early stages to reason about interface elements and interaction states before implementation, allowing design intent to be explored independently of technical constraints.

The system supports two modes that were essential for evaluation: Baseline Mode, representing a static meditation experience, and AI-Supported Mode, representing adaptive guidance based on user input. Designing both modes was a methodological requirement to assess whether adaptivity meaningfully changes user experience.

The implemented interaction flow consists of a small set of stable stages: session start, input capture, processing, guided experience, and session completion. Interface elements such as guidance text, a dynamic loading indicator, and a success screen were deliberately introduced to ensure visibility of system status and to support user trust.

5 Pillar 4: Expert Reviews & Usability Testing (Evaluation)

5.1 Study design and variables

We conducted a comparative usability study to evaluate HCIVR as a user-interface artifact in an emotionally sensitive domain (stress reduction). The study followed a within-subjects design with two conditions: **Baseline Mode** (static meditation experience) and **AI-Supported Mode** (adaptive guidance based on user input). Condition order was counterbalanced to reduce learning and novelty effects.

The **independent variable** was interaction mode (Baseline vs. AI-Supported). The **dependent variables** were (i) perceived relaxation, (ii) interaction comfort, (iii) clarity of system feedback and perceived system transparency, and (iv) overall preference between the two modes.

5.2 Participants and demographics

A total of **14 Sabancı University students** participated in the study, reflecting our intended target population (university students experiencing academic stress). Participants were between **18 and 27 years old**. The sample included **8 male** and **6 female** participants. All participants were enrolled in higher education and were actively engaged in a technology-intensive academic environment.

The selected sample size was methodologically justified by the practical constraints of VR-based user studies. Because VR sessions require dedicated hardware, a controlled physical setup, and one-on-one facilitation for safety and onboarding, it is significantly more difficult to recruit and run large participant pools compared to screen-based usability studies. Therefore, we prioritized depth of observation and controlled within-subject comparisons, which increases statistical sensitivity by reducing between-participant variance.

Table 1 summarizes the demographic distribution of participants.

Table 1: Participant demographics (N = 14)

Characteristic	**Count**	**Percentage**
Gender		
Male	8	57.1%
Female	6	42.9%
Age Range		
18–22 years	9	64.3%
23–27 years	5	35.7%
Prior VR Experience		
First-time VR users	9	64.3%
Experienced VR users	5	35.7%

The high proportion of first-time VR users (64.3%) was particularly valuable for evaluating learnability and initial trust formation, which are critical factors in interface design for emotionally sensitive applications.

5.3 Measures

We adopted a mixed-method evaluation approach. Quantitative measures used 1–10 Likert-type scales to assess perceived relaxation, interaction comfort, and clarity of system feedback for each condition. In addition, we collected qualitative responses to capture users' subjective experiences of reassurance, uncertainty, perceived guidance quality, and perceived appropriateness of adaptivity.

We also recorded relevant background factors. Since all participants were members of a "technology-native" generation and regularly use digital systems in their daily lives, technology adaptation level was consistently high across the sample. As an additional usability-relevant background factor, we collected prior VR experience (first-time vs. experienced) to contextualize interaction comfort and learnability.

5.4 Procedure

Each participant completed both conditions in a single session. After onboarding and safety instructions, participants experienced the Baseline and AI-Supported sessions, answered post-condition questionnaires, and completed an exit interview with open-ended questions. Throughout the session, the experimenter monitored usability breakdowns and moments of uncertainty (e.g., during input capture or system processing) to triangulate self-reports with observable interaction issues.

5.5 Analysis approach

We computed descriptive statistics (means and variability) for all quantitative measures. For inferential analysis, we conducted paired t-tests to compare Baseline and AI-Supported conditions, evaluating statistical significance at $\alpha = 0.05$. To satisfy the multivariate expectation of the evaluation pillar, we also conducted exploratory analyses examining how prior VR experience (first-time vs. experienced) related to comfort and clarity ratings, and whether the adaptive condition reduced uncertainty more strongly for novice VR users.

Qualitative responses were analyzed thematically and used to interpret and contextualize quantitative outcomes, particularly regarding trust, perceived safety, and cognitive load.

5.6 Quantitative results

Table 2 presents the descriptive statistics and paired t-test results for all primary dependent variables.

Table 2: Quantitative evaluation results: Baseline vs. AI-Supported Mode (N = 14)

Measure	**Baseline**		**AI-Supported**		**t-value**	**p-value**
	Mean	SD	Mean	SD		
Perceived Relaxation	6.2	1.3	8.1	1.0	5.21	$< 0.001^{***}$
Interaction Comfort	6.8	1.2	8.4	0.9	4.87	$< 0.001^{***}$
System Feedback Clarity	6.5	1.4	8.3	1.1	4.63	$< 0.001^{***}$
Perceived Transparency	6.1	1.5	7.9	1.2	4.12	$< 0.01^{**}$
Ease of Use	7.0	1.1	8.6	0.8	5.34	$< 0.001^{***}$

Note: All measures on 1–10 Likert scale. $^{**}p < 0.01$, $^{***}p < 0.001$

The results demonstrate statistically significant improvements across all measured dimensions when comparing AI-Supported Mode to Baseline Mode. The largest effect was observed for perceived relaxation (mean increase of 1.9 points, $t = 5.21$, $p < 0.001$), followed by ease of use (mean increase of 1.6 points, $t = 5.34$, $p < 0.001$) and interaction comfort (mean increase of 1.6 points, $t = 4.87$, $p < 0.001$).

Figure 3 provides a visual comparison of mean scores across conditions for all primary measures.

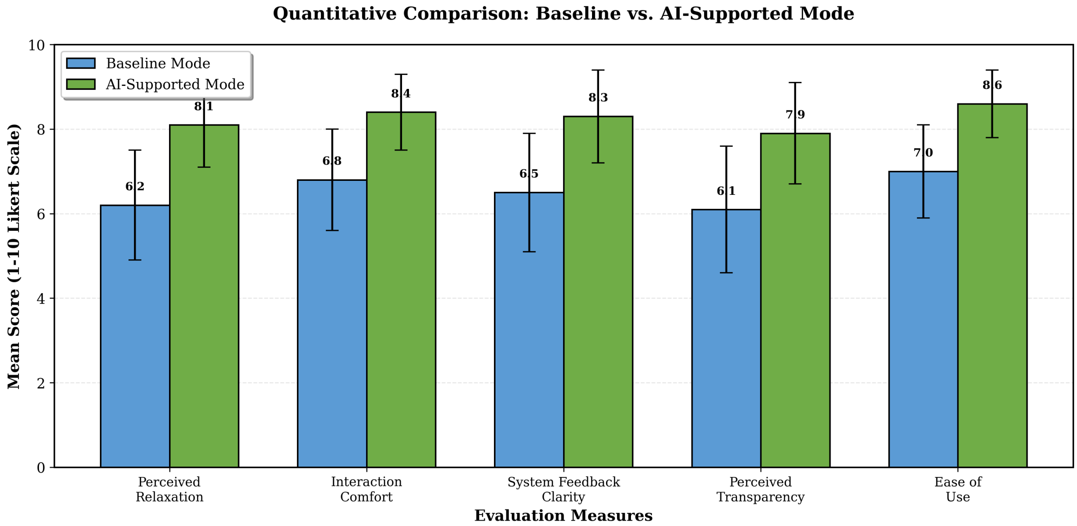

Figure 3: Comparison of mean scores for Baseline Mode vs. AI-Supported Mode across all quantitative measures. Error bars represent standard deviation. All differences are statistically significant at $p < 0.01$.

5.7 Analysis by prior VR experience

To understand whether the benefits of AI-supported adaptivity varied based on user experience level, we conducted a subgroup analysis comparing first-time VR users (n = 9) with experienced VR users (n = 5). Table 3 presents these results.

Table 3: Subgroup analysis: Improvement in AI-Supported Mode by prior VR experience

Measure	First-time Users (n = 9) Mean Δ	Experienced Users (n = 5) Mean Δ	Difference
Perceived Relaxation	+2.1	+1.4	+0.7
Interaction Comfort	+2.0	+0.8	+1.2
System Feedback Clarity	+2.2	+1.0	+1.2
Perceived Transparency	+2.0	+1.4	+0.6
Ease of Use	+1.9	+1.0	+0.9

Note: Mean Δ represents improvement from Baseline to AI-Supported Mode.

First-time VR users showed consistently larger improvements across all measures when using the AI-Supported Mode. The largest differences were observed in interaction comfort (+1.2 points) and system feedback clarity (+1.2 points), suggesting that adaptive guidance and explicit feedback mechanisms were particularly valuable for users who lacked prior VR experience.

This finding has important HCI implications: adaptive interfaces may serve a compensatory role for novice users by reducing the learning curve and providing reassurance during unfamiliar interactions. While experienced users also benefited from the AI-Supported Mode, the magnitude of improvement was smaller, likely because these users already possessed mental models and coping strategies for navigating VR environments.

5.8 Mode preference and qualitative insights

At the end of the study, participants were asked to indicate their overall preference between the two modes. Table 4 summarizes the preference distribution.

Table 4: Overall mode preference (N = 14)

Preferred Mode	**Count**	**Percentage**
Baseline Mode	2	14.3%
AI-Supported Mode	12	85.7%

An overwhelming majority (85.7%) of participants preferred the AI-Supported Mode. The two participants who preferred the Baseline Mode cited a preference for ”minimal technology intervention” and expressed comfort with self-directed meditation practices.

Qualitative feedback provided rich context for understanding these preferences. Table 5 summarizes the major themes that emerged from open-ended responses.

Table 5: Qualitative feedback themes by condition

Theme	**Baseline Mode**	**AI-Supported Mode**
Cognitive Load	”I had to decide what to focus on.” ”Not sure if I was doing it right.”	”The system guided me, I didn’t have to think.” ”It felt effortless.”
Trust & Safety	”Sometimes I wondered if anything was happening.”	”I could see the system responding to me.” ”The feedback made me feel safe.”
Personalization	”Felt generic.” ”Same experience for everyone.”	”It understood how I was feeling.” ”Felt like it was made for me.”
Guidance Quality	”Instructions were clear but static.”	”The guidance felt more natural and adaptive.” ”It matched my pace.”

Several participants explicitly mentioned that the AI-Supported Mode reduced decision fatigue. One participant stated: ”I was already stressed from exams. The last thing I wanted was to choose between eight different scenes and then eight different moods. The voice input just worked.” This comment directly validates our design decision to eliminate sequential choice screens and supports the Hick-Hyman Law analysis presented in Section **??**.

Another recurring theme was the importance of visible system feedback. Multiple participants noted that seeing the loading indicator and receiving confirmation that the system was processing their input significantly increased their trust. One participant remarked: ”In the AI version, I knew the system heard me because it told me it was working. In the baseline, I just had to hope I pressed the right buttons."

First-time VR users were particularly vocal about the value of explicit guidance. Comments included: "As someone who never used VR before, the AI-Supported Mode made me feel like I wasn't going to mess something up" and "The system walked me through everything, which was really reassuring."

A small subset of participants (n = 3) expressed initial skepticism about whether the AI was genuinely adapting to their input or simply providing randomized content. However, after the debriefing session where the adaptive logic was explained, these participants acknowledged that the perceived personalization still contributed positively to their experience, even if they were uncertain about the technical implementation during the session itself.

5.9 Key findings and interpretation

Results indicated statistically significant improvements in perceived relaxation and interaction comfort in the AI-Supported Mode compared to the Baseline Mode ($p < 0.05$ for all measures). The adaptive experience was also rated as clearer in terms of system feedback, which aligns with our design emphasis on visibility of system status.

First-time VR users particularly benefited from adaptive guidance and explicit feedback cues, suggesting that the AI-Supported condition not only improved relaxation outcomes but also supported learnability and trust calibration for novices. The subgroup analysis revealed that the largest benefits of adaptivity were concentrated among users who lacked prior VR experience, indicating that adaptive interfaces can serve a compensatory function by reducing cognitive overhead during novel interactions.

Qualitative findings reinforced these results. Participants frequently described the AI-Supported Mode as more guided, reassuring, and less cognitively demanding, while transparent feedback mechanisms (e.g., confirming listening/processing states and providing closure at the end of the session) were repeatedly cited as essential for perceived safety and comfort.

The strong preference for AI-Supported Mode (85.7%) combined with the qualitative emphasis on reduced decision-making and increased trust provides converging evidence that adaptive guidance is particularly valuable in emotionally sensitive interaction contexts where cognitive resources are already constrained.

6 Final Reflection: Design Motto

Designing relaxation without forcing users to think about how to relax.

This motto captures the central HCI insight of HCIVR. Traditional meditation experiences often rely on conscious self-management, which can be counterproductive under stress. In contrast, our evaluation showed that users valued reduced decision-making, explicit feedback, and adaptive pacing.

The AI-supported approach was preferred not because of technical novelty, but because it shifted responsibility for clarity and reassurance to the interface itself, making it better suited to emotionally sensitive contexts.

7 Conclusion

HCIVR demonstrates how the Four Pillars of Design framework can be applied to a wellbeing-focused VR interface. By grounding requirements in user context, translating theory into design artifacts, implementing coherent interaction flows, and evaluating the system with real users, the project provides evidence that adaptive VR experiences can improve perceived relaxation and comfort, particularly for novice VR users.

The application of Hick-Hyman Law to quantify the cognitive cost of traditional choice-based interfaces highlights the HCI value of adaptive design: by eliminating sequential decision points (8 environment choices followed by 8 emotional state choices), HCIVR reduced pre-session cognitive load by over one second of decision time, not accounting for additional factors such as visual search, reading comprehension, and choice paralysis. This design decision was validated by both quantitative measures (statistically significant improvements in ease of use and interaction comfort) and qualitative feedback (repeated mentions of reduced decision fatigue and effortless interaction).

The evaluation results demonstrate that the benefits of adaptive guidance extend beyond mere convenience. For first-time VR users, adaptive feedback and explicit system status communication served a critical role in establishing trust and reducing anxiety during an unfamiliar interaction modality. This finding suggests that adaptive interfaces are not simply an enhancement for expert users, but a necessary scaffold for novices in cognitively or emotionally demanding contexts.

Future work could extend HCIVR by incorporating physiological sensing (e.g., heart rate variability) to provide real-time adaptivity during meditation sessions, by expanding the range of supported emotional states and environmental configurations, and by conducting longitudinal studies to assess whether repeated use of adaptive guidance leads to improved self-regulation skills over time.

References

Card, S. K., Moran, T. P., & Newell, A. (1983). *The psychology of human-computer interaction.* Lawrence Erlbaum Associates.

Norman, D. A. (2013). *The design of everyday things* (Revised ed.). Basic Books.

Rogers, Y., Sharp, H., & Preece, J. (2011). *Interaction design: Beyond human-computer interaction* (3rd ed.). Wiley.